Data Visualization 2026/27 | M:DS, FCUP

Project. Weekly submissions and checkpoints

Why weekly submissions

The project is built in stages that follow the tools taught in the course: ggplot2 for the exploration, plotly for the interaction, Shiny for the dashboard. Each week the group adds something to the previous week's version, and the course keeps a record of that growth. The record has two uses: feedback at four checkpoints, and a final presentation in which the time goes entirely to discussing your choices.

What to do every week

1. The project folder (ZIP)

Before Friday 23:30, zip the whole project folder as it stands and submit it in the Moodle activity "`Project submission N (ZIP)`". The ZIP may be submitted at any time during the week and replaced as many times as you want. The last ZIP uploaded before the deadline is the one that counts.

- The folder should be cumulative: each ZIP contains everything the group has so far (scripts, figures, app, documents), including the work of earlier weeks.
- The **dataset file should stay out of the ZIP**. It is large and identical for everyone. Keep it in a `data/` folder that you do not include, or even located outside the project folder.
- Figure files should keep their names from week to week. If `fig/complaints_by_hour.png` is improved, it stays `fig/complaints_by_hour.png`. Only new figures get new names.
- Scripts may be `.R`, `.Rmd` or `.qmd` files. A Shiny app is an `app.R` file, or `ui.R` and `server.R`.
- A week with no submission, or with a ZIP identical to the previous one, is recorded as a week without progress.
- Submissions between Friday 23:30 and Friday 23:59 are accepted and marked as late. After that the activity closes.

A simple structure that works:

```
project/
  R/        scripts (.R, .Rmd or .qmd)
  fig/      figures produced by the scripts
  app/      the Shiny app, from checkpoint 4
  docs/     proposal, article, slides (PDF)
  data/     the data file, NOT to be included in the ZIP
```

2. The progress check (one minute)

In the Moodle activity "`Progress check N`", one member of the group (decide who) ticks what the group did during the week. There are no text fields and the answers have no grade. They are read together with the ZIP.

Question	Options
Group	your group
This week the group... (tick all that apply)	explored new variables of the data, created new figures, modified existing figures, cleaned or transformed data, formulated or tested hypotheses, discarded a hypothesis or a figure, made figures interactive (plotly), worked on the Shiny app, wrote text (proposal, article, slides), no progress this week
Tools used this week	ggplot2, dplyr or tidyr, plotly, Shiny, other…
AI tools this week	none, to explain code or error messages, to write or fix code, to improve text, to suggest analyses or figures, other
Main difficulty this week (one answer)	the data, the code, interpreting results, time, none

Calendar

Classes are on Mondays and the deadlines are on the following Friday at 23:30.

Week	Class (Monday)	Deadline (Friday 23:30)	Submission and checkpoint
1	26 October	30 October	ZIP 1 and progress check 1
2	2 November	6 November	ZIP 2 and progress check 2. Checkpoint 1: questions and exploration
3	9 November	13 November	ZIP 3 and progress check 3
4	16 November	20 November	ZIP 4 and progress check 4. Checkpoint 2: hypotheses
5	23 November	27 November	ZIP 5 and progress check 5. Checkpoint 3: interaction
6	30 November	4 December	ZIP 6 and progress check 6
7	7 December	11 December	ZIP 7 and progress check 7. Checkpoint 4: dashboard and article
	14 December		Final presentation, with demo

The four checkpoints

A checkpoint is the ZIP of that week with some required content. How many figures, hypotheses or interactive views the group produces is the group's decision. Each checkpoint receives written feedback before the next class.

Checkpoint	What the ZIP must contain	Link to the previous checkpoint
1st, 6 November	The research questions the group chose. The exploration in ggplot2 (scripts and figures). The project proposal (PPA, 4 pages, PDF)	Starting point. The questions stay as the reference for the rest of the project
2nd, 20 November	The hypotheses derived from the questions, each with the figure that supports or refutes it. The hypotheses the group discarded, and why (a short text in any format)	Same questions as checkpoint 1. Figures that evolve from those of checkpoint 1

Checkpoint	What the ZIP must contain	Link to the previous checkpoint
3rd, 27 November	The figures the group chose to make interactive with plotly, each with one sentence on what the interaction adds	Figures coming from checkpoint 2
4th, 11 December	The Shiny app (code) and the article (PD, 6 pages, IEEE format, PDF)	The app integrates the figures of checkpoint 3. The article follows the hypotheses of checkpoint 2

If the group changes its questions or its audience after checkpoint 1, that is allowed. The reasons should appear in the proposal or in the article, so that the change can be discussed at the presentation.

What happens with your submissions

The teacher compares consecutive versions of each group's folder: how much code is new each week, which figures persist and evolve, and whether each checkpoint builds on the earlier weeks. The progress checks are read alongside the ZIPs.

The weekly ZIPs have no grade of their own. The proposal (PPA), the article (PD) and the final presentation (FP) are graded as announced in the first lecture. At the final presentation the time goes to your choices: why these questions, why this figure, why this interaction, etc. Make sure every member of the group can explain every figure the group kept.

Before you submit on Friday

- The ZIP contains the whole project folder as it stands today.
- The data file is not inside.
- Figure files kept their names from last week, and new figures have new names.
- In checkpoint weeks, the required content is in the folder (the proposal or the article as PDF).
- One member has filled in the progress check.

Questions you may have (FAQ):

Can we submit earlier than Friday? Yes, at any time during the week. Only the last version before the deadline is kept.

Do we need Git? No. Git is useful for the group's own work, but the course uses only the Moodle submissions.

We reorganised our scripts. Is that a problem? No. Moving code between files is fine. Keep the figure names.

We made little progress this week. Submit the ZIP anyway and tick "no progress this week" in the progress check. The record simply shows a quiet week.

Can we use AI tools? Partially: only to create a portion of code, not for dataset inspection or for batch creation of figures. Declare the use in the progress check. The project remains yours: at the presentation you will be asked about your choices, so keep only what you can explain.

The ZIP is too large to upload. Almost always the data file or an `.RData` file is inside the ZIP. Remove them and submit again.